



SOP — Build the "10-Minute" Platformer in Godot 4

The step-by-step procedure for Eli Cuaycong's famous 10-minute platformer video — **already translated to Godot 4**, the version we use at camp. Each step links to the exact moment in the video.

Video: ["Make your first 2D platformer game IN JUST 10 MINUTES"](#) by Eli Cuaycong · Free sprites: [hajileee's Fantasy Characters Pack \(itch.io\)](#)

⚠ Read this first — the video is OLD Godot (3), we use Godot 4

The video is great, but it was made in Godot 3. Our camp computers run **Godot 4**, where some names changed. **This SOP is already written the Godot 4 way** — follow the paper, watch the video for the ideas. When the video says the old word, think the new one:

VIDEO SAYS (GODOT 3)	WE USE (GODOT 4)
KinematicBody2D	CharacterBody2D
Sprite	Sprite2D
move_and_slide(motion, UP)	velocity + move_and_slide()
"Convert scene to TileMap (.tres)"	TileMapLayer node + TileSet resource
Shape "Extents"	Shape "Size"
Camera "Current" + "Smoothing"	Camera "Enabled" + "Position Smoothing"

1 Get art + make pixels crisp

▶ 0:26

TOP BAR

1. Download the free sprite pack (link above) and drag the PNGs into your **Files** panel.
2. Pixel art looks blurry at first. Fix it once for the whole project: **Project** → **Project Settings** → **Rendering** → **Textures** → **Default Texture Filter** → **Nearest**.

Video difference: the video re-imports each sprite with a "2D Pixel" preset — Godot 4's one setting above does it for everything.

2 Make the tiles (the ground pieces)

▶ 0:14

■ PARTS LIST

1. New scene → root **Node2D**, rename it **World**.
2. Add child → **TileMapLayer**.
3. Click it. In **Settings: Tile Set** → **New TileSet**, and set **tile size** to match your art (the video's tiles are 64×64).
4. Open the **TileSet** panel (bottom of the screen) → drag your tile PNG in → Godot slices it into tiles automatically.
5. Still in TileSet: **Physics Layers** → **Add Element**, then paint the physics shape on each tile. (This is what makes tiles solid — the video's StaticBody2D + collision step, done the new way.)

3 Draw your level

▶ 1:25

■ STAGE

1. Select the **TileMapLayer** → the **TileMap** panel opens at the bottom.
2. Pick a tile, then **paint your level** in the Stage like a coloring book. Floors, walls, platforms — your world, your rules.

4 Build the Player (the family of 4)

▶ 1:49

■ PARTS LIST

1. Under **World**, add a **CharacterBody2D** → rename it **Player**. (Video says "KinematicBody2D" — same thing, new name.)
2. Give Player three children: **Sprite2D** (her picture) + **CollisionShape2D** (her bump-shape) + **Camera2D** (the eye that follows her).
3. On **CollisionShape2D** → Shape → **New CapsuleShape2D** → drag the handles until it hugs the character.
4. On **Camera2D** → set **Zoom** to taste (2×2 works for 32px art) → turn on **Position Smoothing**.

5 Load the sprite sheet frames

▶ 2:47

■ SETTINGS

1. Click **Sprite2D** → **Texture** → **Load** → pick the character sheet.
2. In **Settings** → **Animation**: set **Hframes** = **8** and **Vframes** = **6** (that's this pack's grid — one pose per 32×32 box).
3. Leave **Frame** = **0** for now.

6 Teach the keys (Input Map)

▶ 3:50

■ TOP BAR

1. **Project** → **Project Settings** → **Input Map**.
2. Add three actions, spelled exactly: **left** · **right** · **jump**.
3. Give each a key: **A**, **D**, and **Space** (add the arrow keys too if you like).

Spelling matters: the code below looks these names up letter-for-letter.

7 The brain, part 1 — walk & fall

▶ 3:29

■ PARTS LIST → SCRIPT

Right-click **Player** → **Attach Script** → **Create**. The full finished script is at the bottom of this page — this step and the next two build it up the same order the video does:

Walk + gravity (the video's first chunk, in Godot 4)

```
extends CharacterBody2D

const GRAVITY = 1200.0      # pulls her down [4:43]
const MAX_FALL_SPEED = 400.0 # falling speed cap [5:03]
const MAX_SPEED = 250.0     # top running speed

func _physics_process(delta):
    if Input.is_action_pressed("right"):    # [4:15]
        velocity.x = MAX_SPEED
    elif Input.is_action_pressed("left"):
        velocity.x = -MAX_SPEED
    else:
        velocity.x = 0

    velocity.y += GRAVITY * delta           # gravity every tick
    velocity.y = min(velocity.y, MAX_FALL_SPEED)

    move_and_slide()                        # do the move [5:26]
```

Video difference: Godot 3 needed `move_and_slide(motion, UP)`.

Godot 4's `CharacterBody2D` has velocity built in — call `move_and_slide()` with nothing.

8 The brain, part 2 — jump

▶ 5:38

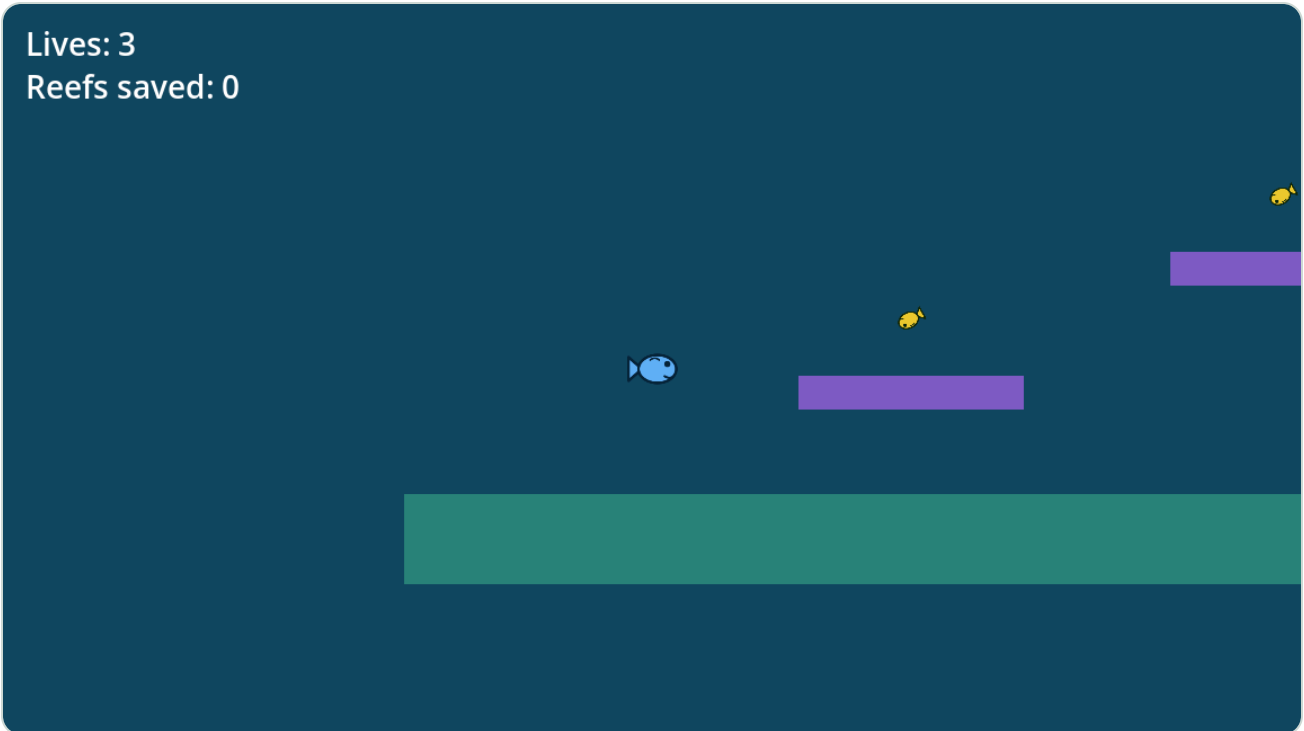
■ SCRIPT

Add inside `_physics_process`, before `move_and_slide()`

```
const JUMP_FORCE = -450.0          # negative = UP!  (top of script)

if Input.is_action_just_pressed("jump") and is_on_floor():
    velocity.y = JUMP_FORCE  # only when standing - "ain't flappy bird" [6:05]
```

Lives: 3
Reefs saved: 0



Real screenshot from our Godot 4 build — mid-jump. This is what success looks like at this step.

9 The brain, part 3 — make it feel good

▶ 6:16

■ SCRIPT

This is the video's secret sauce — instead of instant speed, she **speeds up** and **glides to a stop**:

Replace the walk lines from step 7 with these

```
const ACCELERATION = 40.0          # how fast she speeds up [6:31]

    if Input.is_action_pressed("right"):
        velocity.x += ACCELERATION      # builds up speed
    elif Input.is_action_pressed("left"):
        velocity.x -= ACCELERATION
    else:
        velocity.x = lerp(velocity.x, 0.0, 0.2) # glide to a stop [7:12]

velocity.x = clamp(velocity.x, -MAX_SPEED, MAX_SPEED) # speed cap [6:58]
```

lerp sounds scary but it just means "slide a bit of the way toward zero each tick" — that's the smooth Sonic-style stop.

10 Animate her + press GO 🎉

▶ 7:44

■ PARTS LIST + SCRIPT

1. Add an **AnimationPlayer** as a child of Player.
2. In the **Animation** panel (bottom): make 4 animations — **idle** (loop), **run** (loop), **jump** (2 frames, no loop), **fall** (4 frames, no loop).
3. For each: select **Sprite2D**, and in **Settings → Animation → Frame** click the little **key** ♦ icon once per frame of that pose — this drops keyframes into the timeline. ▶ 8:02
4. Wire it up in the script (this replaces nothing — just add):

Flip + play the right animation [8:43 → 9:28]

```
var facing_right = true
@onready var sprite = $Sprite2D
@onready var anim = $AnimationPlayer

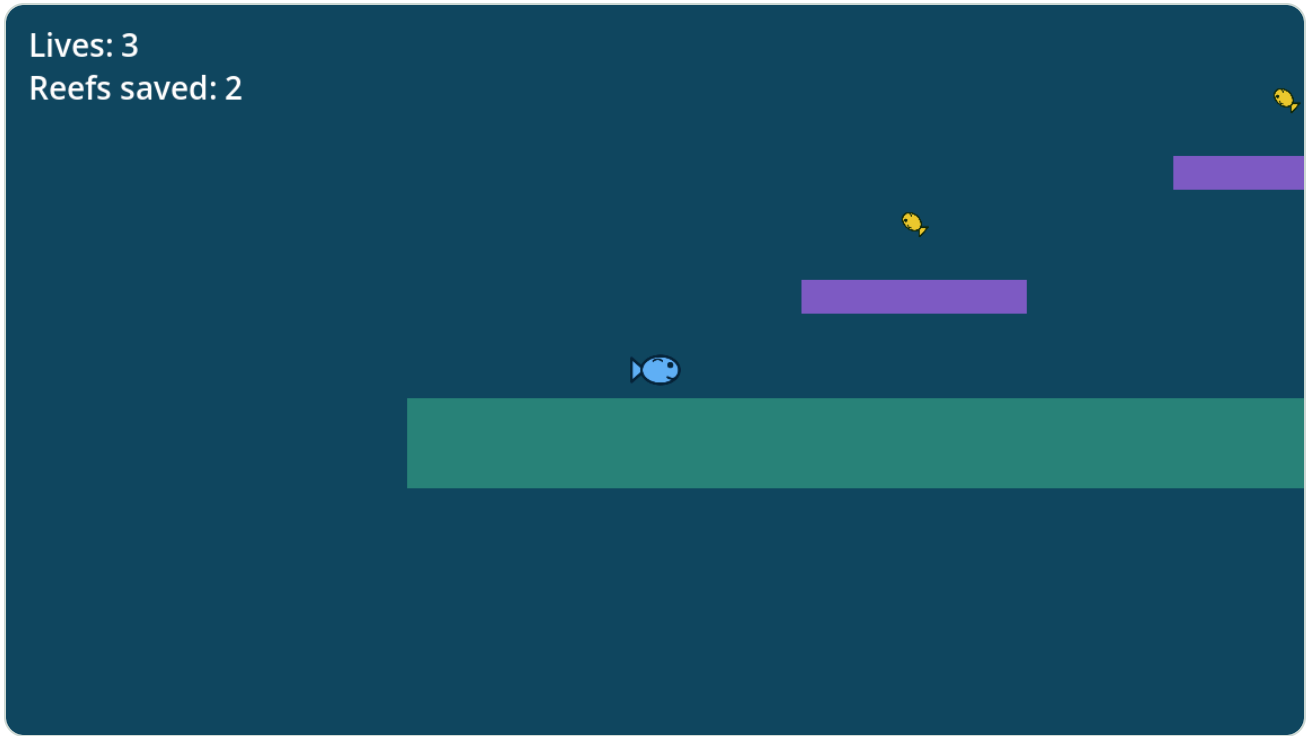
# inside _physics_process:
if Input.is_action_pressed("right"): facing_right = true
elif Input.is_action_pressed("left"): facing_right = false
sprite.flip_h = not facing_right

if is_on_floor():
    if abs(velocity.x) > 5.0: anim.play("run")
    else: anim.play("idle")
else:
    if velocity.y < 0.0: anim.play("jump")
    else: anim.play("fall")
```

5. **Save** (⌘/Ctrl+S) and press ▶ **GO** (F5). Pick **World** as the main scene.

▶ 9:37

Lives: 3
Reefs saved: 2



The finished loop running for real (from our Reef Runner build): the HUD counts what she collects.



The whole brain in one piece (copy-paste safe)

Player.gd — complete, Godot 4

```
extends CharacterBody2D

const GRAVITY = 1200.0
const MAX_FALL_SPEED = 400.0
const MAX_SPEED = 250.0
const ACCELERATION = 40.0
const JUMP_FORCE = -450.0

var facing_right = true
@onready var sprite = $Sprite2D
@onready var anim = $AnimationPlayer

func _physics_process(delta):
    # walk with acceleration
    if Input.is_action_pressed("right"):
        velocity.x += ACCELERATION
        facing_right = true
    elif Input.is_action_pressed("left"):
        velocity.x -= ACCELERATION
        facing_right = false
    else:
        velocity.x = lerp(velocity.x, 0.0, 0.2)
    velocity.x = clamp(velocity.x, -MAX_SPEED, MAX_SPEED)

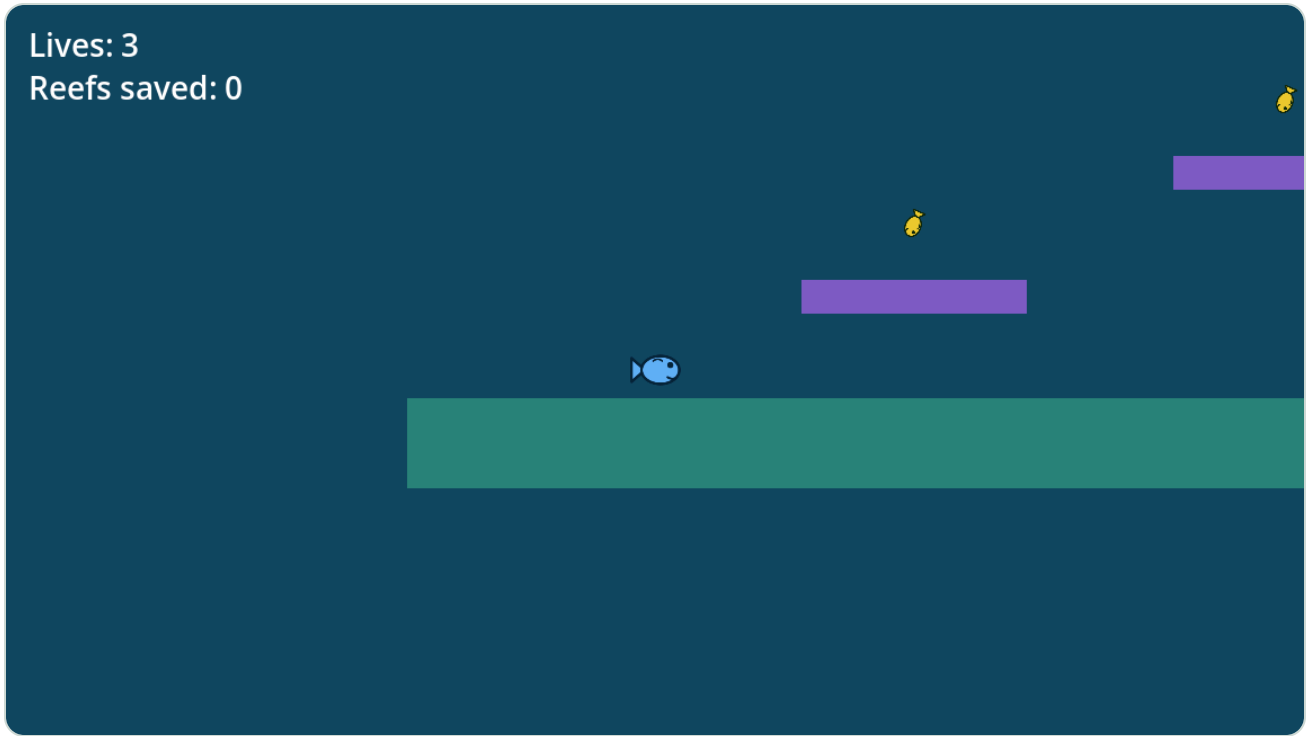
    # gravity + fall cap
    velocity.y += GRAVITY * delta
    velocity.y = min(velocity.y, MAX_FALL_SPEED)

    # jump only from the floor
    if Input.is_action_just_pressed("jump") and is_on_floor():
        velocity.y = JUMP_FORCE

    # face the right way + pick the animation
    sprite.flip_h = not facing_right
    if is_on_floor():
        anim.play("run" if abs(velocity.x) > 5.0 else "idle")
    else:
        anim.play("jump" if velocity.y < 0.0 else "fall")

    move_and_slide()
```

Lives: 3
Reefs saved: 0



Where you start (our Reef Runner Template — same recipe, already built). Compare: same player brain, same platforms, same GO button.

Credits & sources: Steps follow [Eli Cuaycong's video](#) (Godot 3), translated to Godot 4 for the BGC camp. Sprites: [hajileee on itch.io](#) (free; donations unlock enemies). Screenshots are from our own Godot 4.7 build of the Reef Runner Template — not from the video. If a step looks different on your screen, check the Godot 3 → 4 table at the top first.